

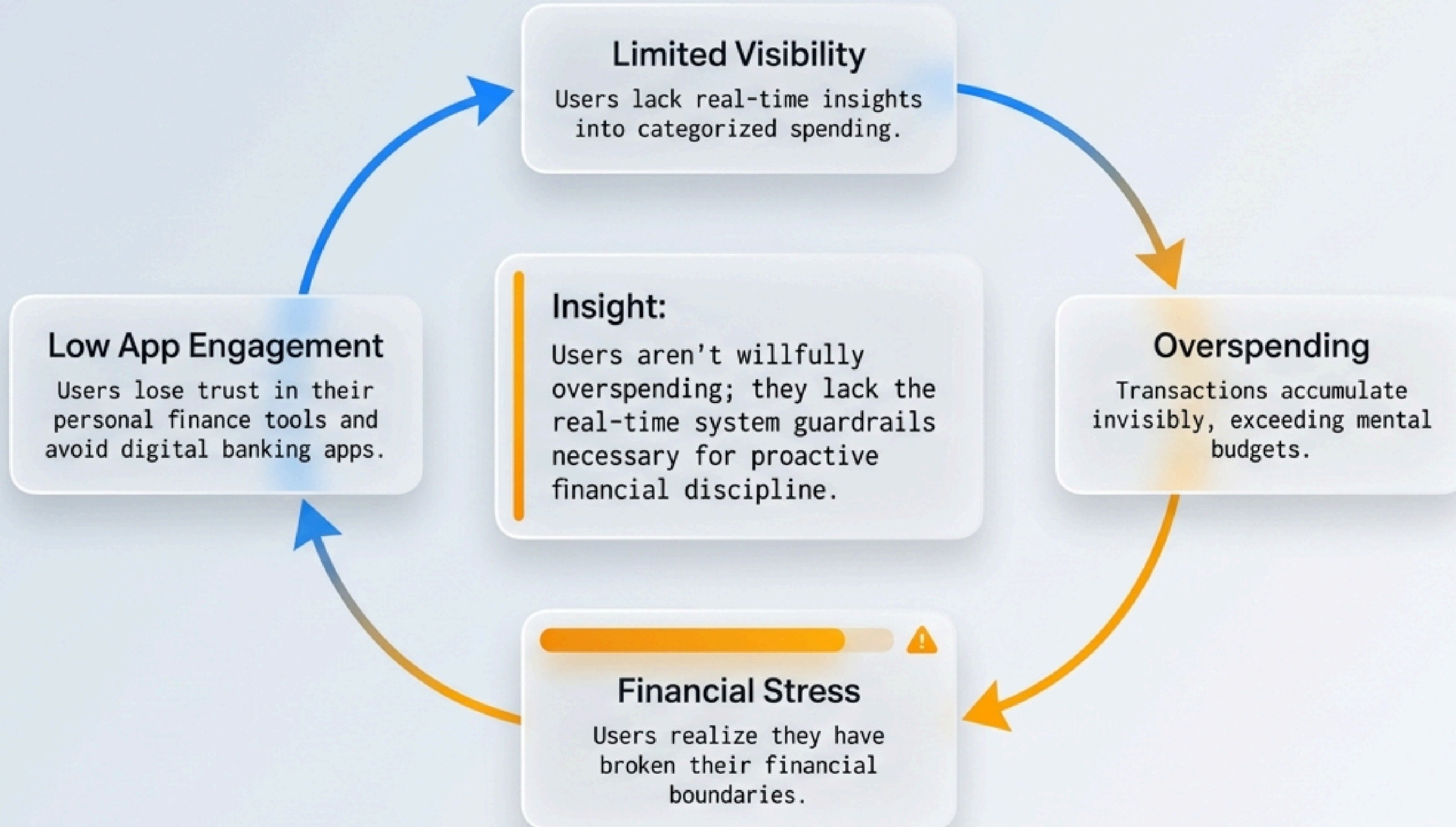
# Smart Expense Tracker: Budget Alert Blueprint



From Business Logic to User Impact



# The Vicious Cycle of Unmonitored Spending





# The Target Audience (Who We Are Building For)



## The Engineer Malcolm

**Financial Footprint:** Multiple accounts including MB Bank Savings and Credit Cards with high transaction volumes.

**Need:** Automated categorization for complex spending.



## The Student Rica

**Financial Footprint:** VPBank Debit Card.

**Need:** Strict, low-friction budget boundaries to manage limited funds.



## The Freelancer Koren

**Financial Footprint:** TPBank Savings & Techcombank.

**Need:** Real-time alerts to manage variable income against fixed entertainment and travel expenses.

Targeting 1,000+ concurrent users across diverse institutional integrations (ACB, MB Bank, VPBank, Techcombank, TPBank).



# Feature Overview & Scope

## In-Scope (The Core Engine)

- ✓ Monthly budget creation by category
- ✓ Continuous budget threshold monitoring
- ✓ In-app alert notifications (triggered at 80%)
- ✓ Alert history logging and display
- ✓ Automatic transaction categorization

## Out-of-Scope (Strict Boundaries)

- ✗ SMS or Email notifications (In-app only)
- ✗ AI-based budget recommendations
- ✗ Investment portfolio or loan tracking
- ✗ Credit scoring management

Synthesis: The focus is strict, automated guardrails—delivering certainty, not AI speculation.



# The User Experience: Frictionless Integration

## US-01: Frictionless Creation.

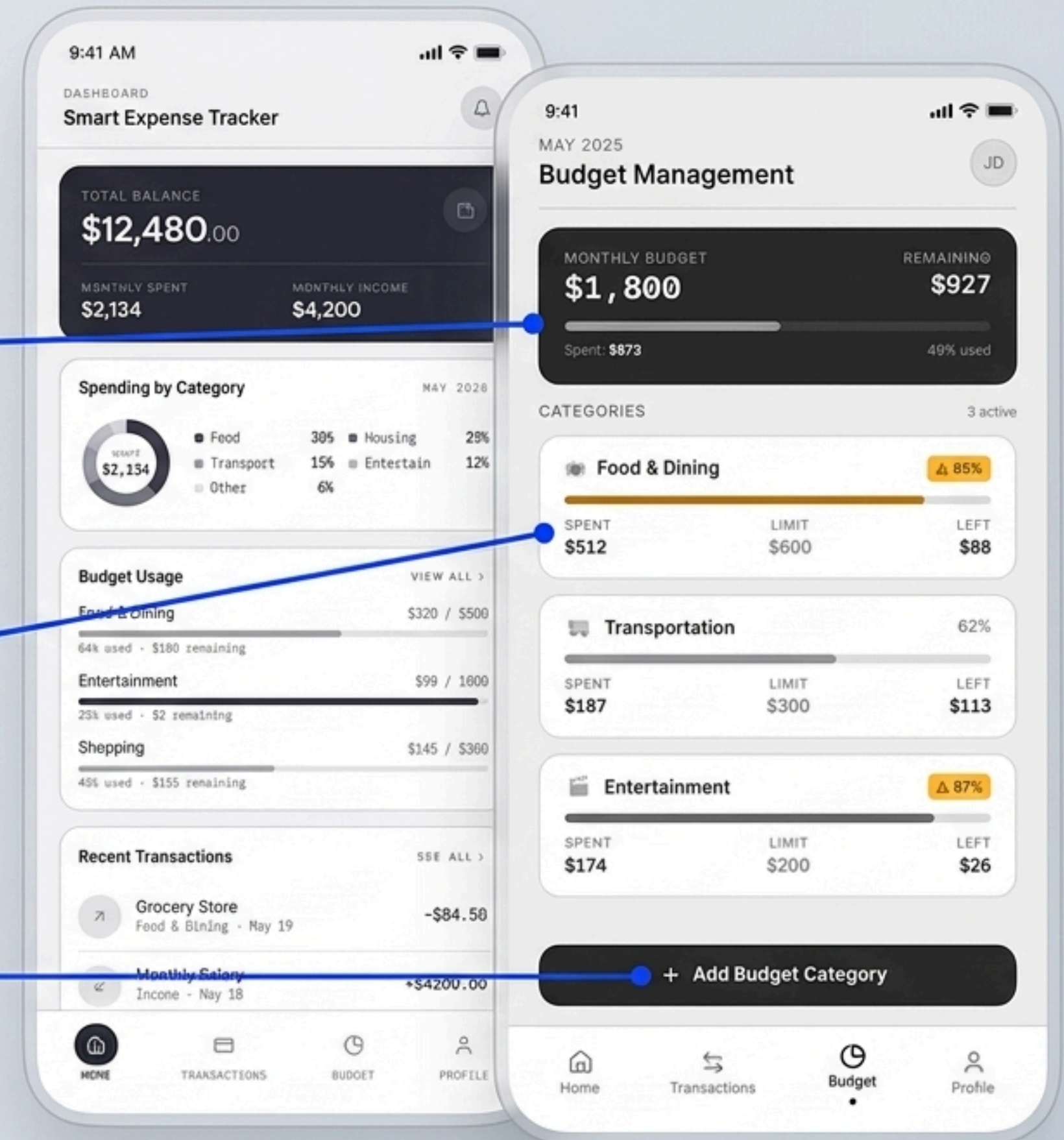
Users define monthly limits per expense category. Requirement NFR-07 dictates this configuration takes < 3 clicks.

## Visual Anchoring.

Real-time consumption bars auto-fill as transactions process, preparing the user for the impending alert.

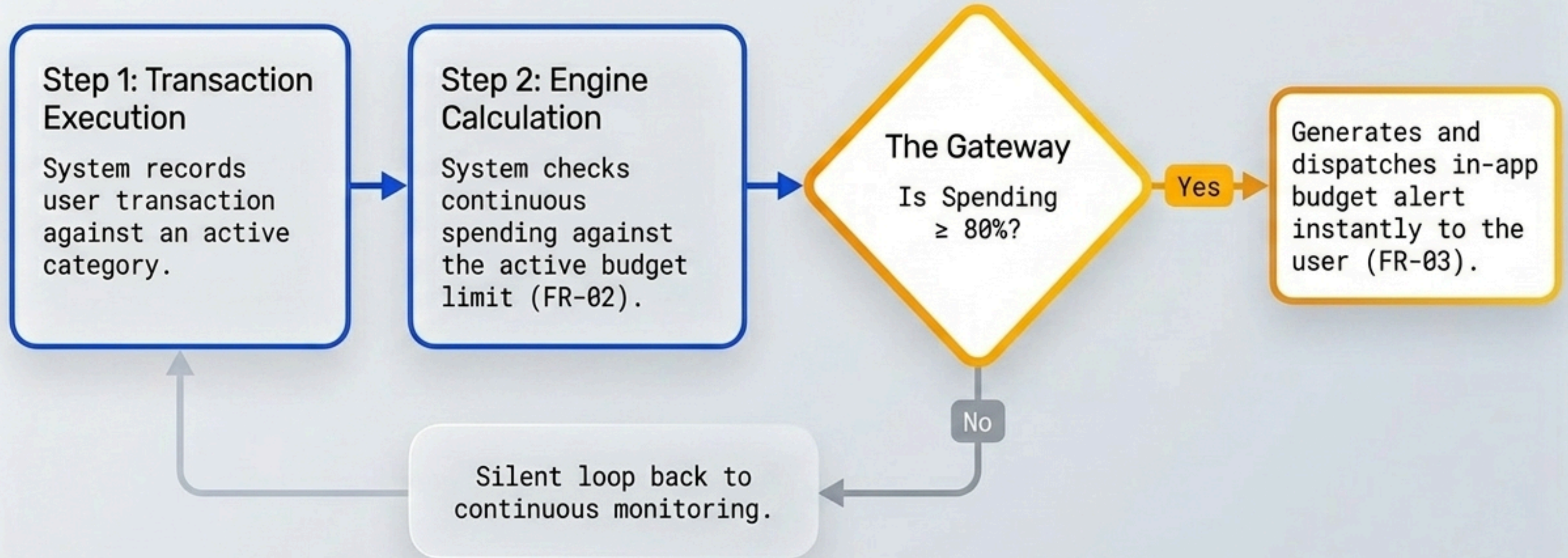
## FR-01: Category Customization.

Extensible architecture allows users to tailor guardrails to specific lifestyle needs.





# The Core Engine: 80% Alert Logic





# System Guardrails Matrix

## Data Input Rules

**Categorization:** Every transaction belongs to exactly one category (BR-02, AL-04).

**Null Budgets:** If no budget exists for a category, no alert is triggered (BR-08).

## Processing Engine Constraints

**The Threshold:** Default alert trigger hardcoded to 80% (AL-02).

**Instant Updates:** Modified budget thresholds apply to the engine immediately (BR-06).

**Spam Prevention:** Absolute block on duplicate alerts for the same category within the same month (BR-07).

## Reset & Delivery Triggers

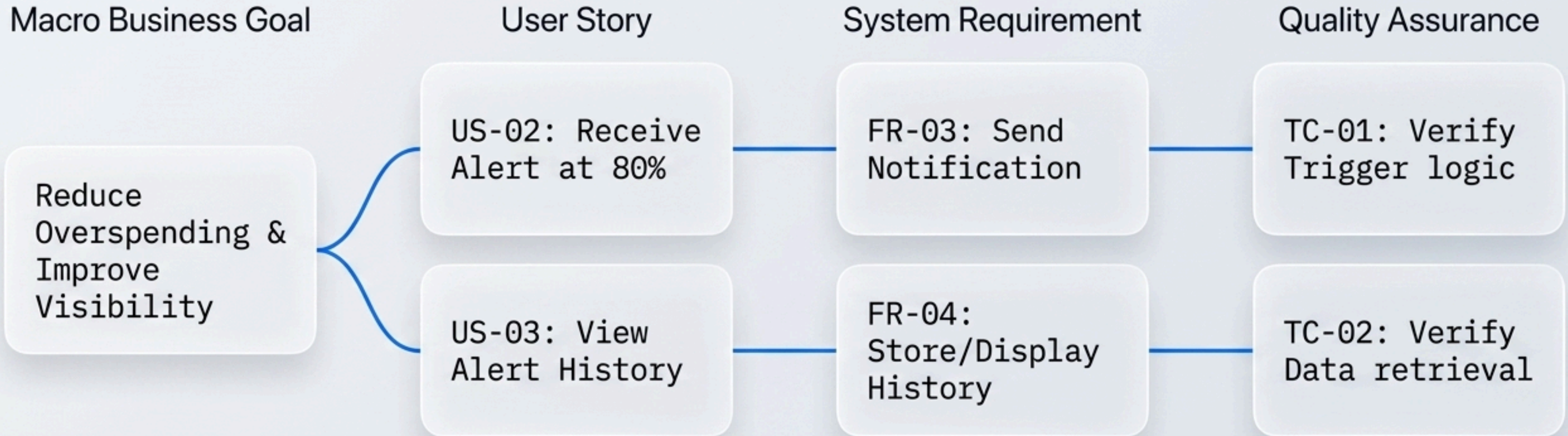
**Temporal Reset:** All counters zero out automatically on the 1st day of the month (BR-03, AL-05).

**Delivery Mechanism:** In-app push only; external channels disabled (AL-03).



# Traceability: Business Goal to Execution

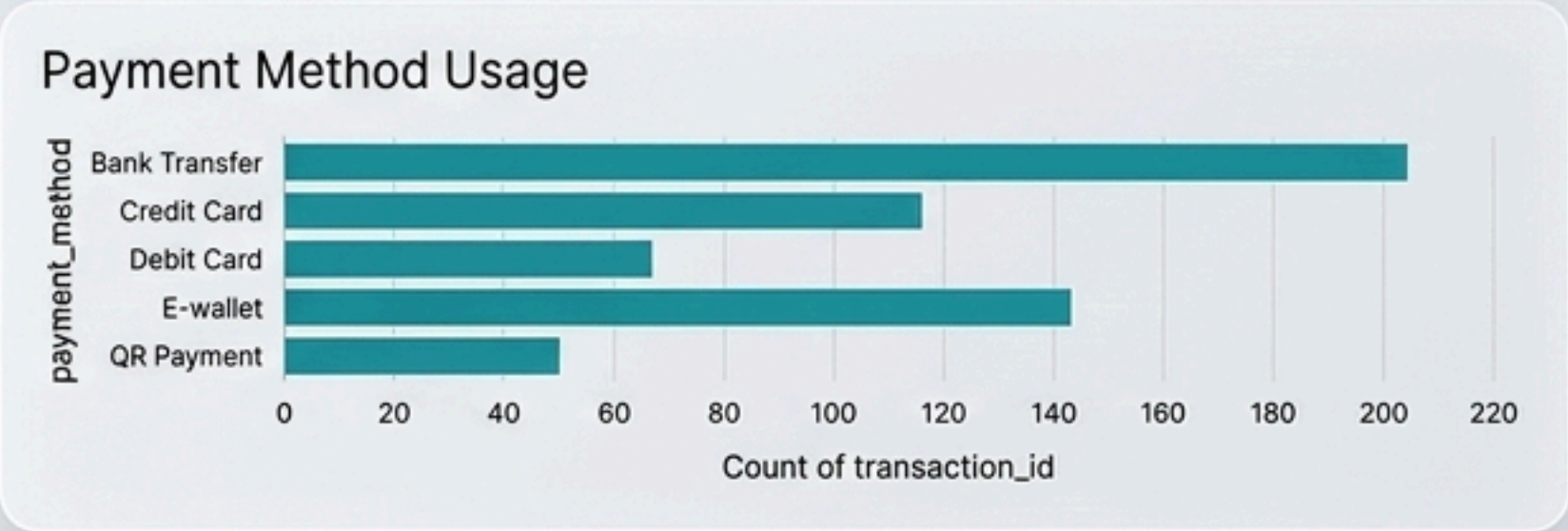
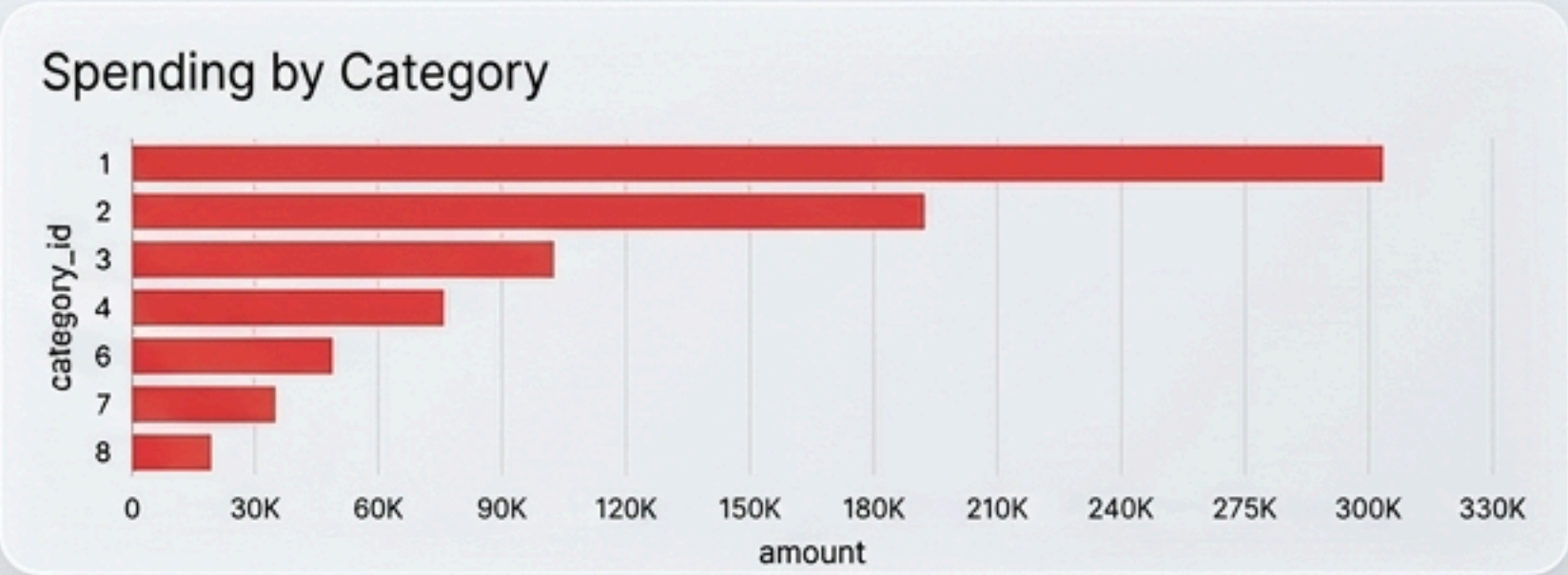
## Impact Cascades



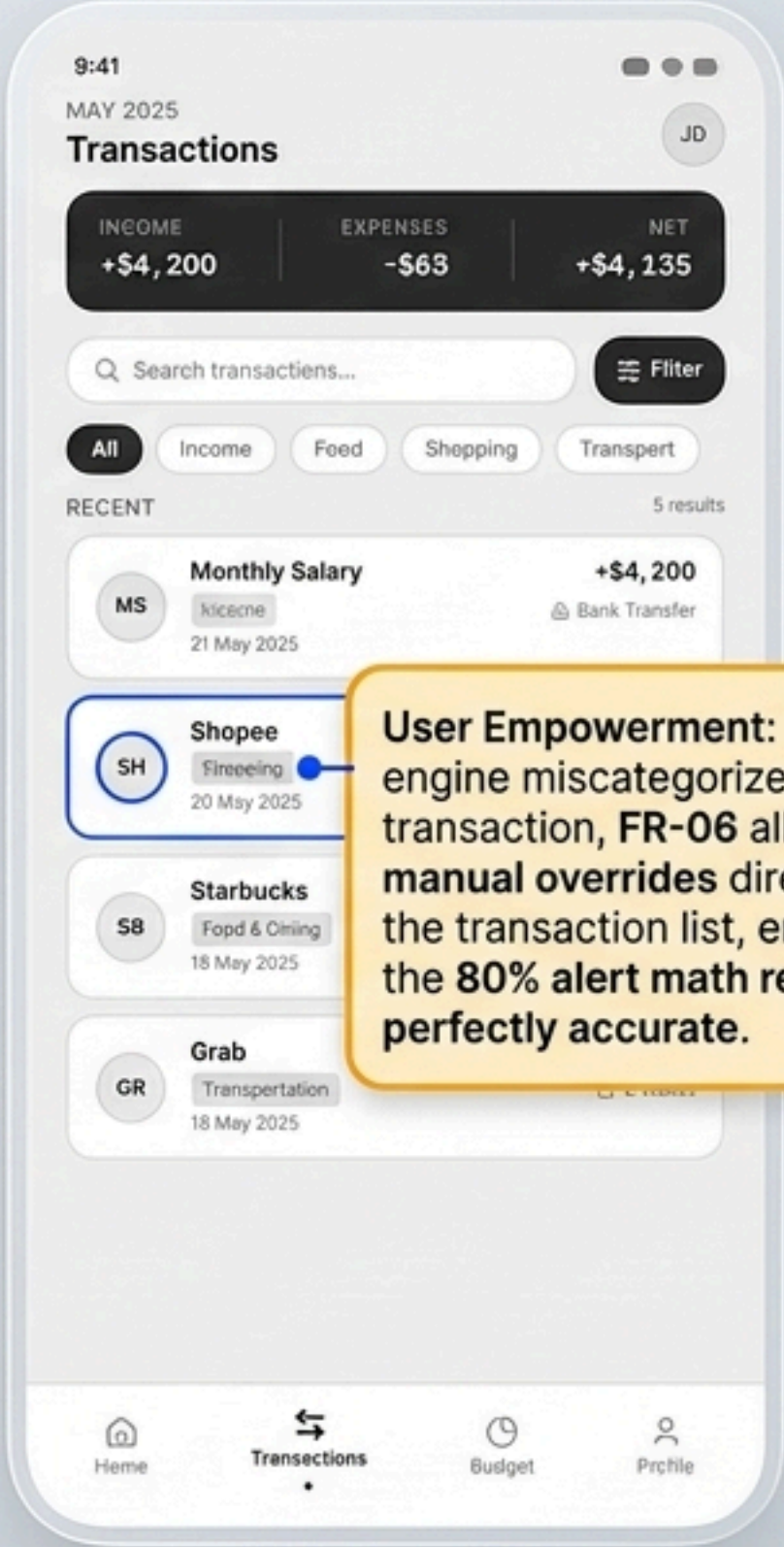
**Absolute Traceability:** From high-level business strategy directly down to unit execution. No orphaned features; no unverified requirements.



# Managing Transactions & Categorization



The **Data Engine**: Transactions flow in via E-Wallet, QR Payment, and Bank Transfer. FR-05 auto-categorizes them based on historical vendor data.





# Non-Functional Guardrails (Technical Standards)

## Performance Dashboard

### < 2 Seconds

**Speed:** Alerts triggered and delivered within 2 seconds of the 80% threshold detection (NFR-01).



### 1,000 Users

**Scale:** Architecture sized to support 1,000 concurrent user sessions flawlessly (NFR-04).



### 99% Uptime

**Reliability:** High availability coupled with a >95% guaranteed alert delivery success rate (NFR-02, NFR-05).



### Zero-Trust Data

**Security:** All sensitive financial configurations and transaction histories are fully encrypted at rest and in transit (NFR-03).





# Success Metrics & Impact

## Future State Impact



Behavioral Impact

### **Reduced Overspending**

Measured by a month-over-month decrease in transactions occurring after an 80% alert is fired.



Tool Adoption

### **Increased Budget Creation**

Measured by the volume of active, manually defined budget limits set across the user base.



Financial Awareness

### **Improved Visibility**

Measured by frequency of user visits to the dashboard and alert history screens.



Platform Health

### **Enhanced Digital Engagement**

Measured by overall app retention and user trust metrics.



# Summary & Next Steps

Pillar 1

## **Strategic Alignment**

Directly solves the root cause of financial stress through real-time visibility, tracing every user story back to a core business goal.

Pillar 2

## **Technical Precision**

Governed by airtight business rules, duplicate-prevention logic, and sub-2-second performance thresholds.

Pillar 3

## **Consumer-Grade UX**

Wraps complex database categorization and threshold mathematics into a frictionless, 3-click interface.

**The Smart Expense Tracker is fully architected. From backend categorization to frontend alert dispatch, the blueprint is ready for execution.**